

NGUYEN GIA HUY

Backend Engineer

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PROFESSIONAL SUMMARY

Backend engineer with nearly three years of experience building transaction-oriented services and integrations. Integrated a Thought Machine Vault Core banking platform with external services and implemented its lending and overdraft transaction logic; built a Golang account migration system on AWS that runs **5x faster**; and built a gRPC, Kafka, MongoDB, and Redis backend serving **1.5K requests/second** at **80–100 ms** with **99.95% uptime**. Works across API contracts, data stores and search, messaging, batch processing, and the gateway and identity layers of a **100+ service** microservices ecosystem.

TECHNICAL SKILLS

Languages:	Golang, Python, SQL
APIs:	gRPC, gRPC-Gateway (REST/JSON), Protocol Buffers, FastAPI
Data Stores:	PostgreSQL, Amazon RDS, MongoDB, Redis, OpenSearch, Elasticsearch, ClickHouse
Messaging & Events:	Kafka, RabbitMQ, event-driven services
Service Infrastructure:	Kong Gateway, Keycloak (RBAC), Docker, Kubernetes, AWS (ECS, Lambda, EKS, RDS)
Observability:	OpenTelemetry, Prometheus, Grafana, ELK, CloudWatch, DataDog

WORK EXPERIENCE

Backend Engineer, Core Banking (Middle Software Engineer) 03/2026 – Present
Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

Integration and Transaction Logic

- Integrated **Thought Machine Vault Core** with external backend services, wiring the core ledger into the bank's operational and transaction workflows.
- Implemented the Lending and Overdraft transaction logic as Python Smart Contracts on Vault Core, and maintained it in production.
- Implemented interest accrual and penalty and late-fee calculation for both products.
- Implemented repayment and settlement processing for both products.

Migration Backend

- Built the account migration backend in **Golang**, shortening a migration from **2.5 hours to 27 minutes (5x)**.
- Ran the migration processing on AWS ECS and Lambda, persisting to **RDS**.

Testing and Deployment

- Rewrote the simulation testing framework in Golang, replacing the Gauge-based framework.
- Gave the transaction logic fast test feedback: simulation tests now finish in **just over 50 seconds instead of 24 minutes**.
- Deployed a new Vault Core version on AWS (EKS, RDS, ECR, ELB) with a CI/CD pipeline for build, test, and deployment.

Software Engineer – Backend & Data Services 04/2025 – 03/2026
Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Migrated the search and analytics backend from **Elasticsearch to OpenSearch**.
- Re-architected data ingestion and indexing during the migration, improving stability, consistency, and query performance.
- Raised batch throughput by **150%** by redesigning pipelines, reworking data access patterns, and running processing steps in parallel.
- Built backend data services on AWS Lambda, ECS, RDS, S3, and Athena that transform large-scale utility data for analytics and operational consumers.
- Managed the services' infrastructure with Terraform and AWS SAM.
- Monitored them with CloudWatch and DataDog for health, failures, and anomalies.

Software Engineer – API Gateway & Identity

04/2024 – 04/2025

VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Operated **Kong Gateway** for a microservices ecosystem of **100+ services**.
- Enforced service-level and network-level access control at the gateway.
- Built multi-tenant **authentication** on **Keycloak** for **20+ customer organizations**.
- Implemented role-based **authorization** that isolates each tenant's access.
- Operated the OpenStack and Cluster API private cloud behind the platform: **50+ production Kubernetes clusters** with automated provisioning, scaling, and zero-downtime upgrades.
- Automated about **80%** of data center provisioning and infrastructure configuration.

Blockchain Developer – Event-Driven Backend

06/2023 – 09/2023

EsolLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Built **event-driven backend services** that capture, process, and synchronize on-chain events into off-chain databases and APIs.
- Integrated them with **5+ blockchain networks** (Ethereum, BSC, Polygon, Aptos, Sui).
- Kept transaction handling consistent and data intact across chains.
- Developed and reviewed **15+ smart contracts** for DeFi and NFT use cases in Solidity and Move.

SELECTED PROJECTS

DattingQ | Software Engineer

Dating platform

- Built the API layer with **gRPC** services exposed as REST/JSON through **gRPC-Gateway**.
- Backed the services with **Kafka**, **MongoDB**, and **Redis**, sustaining up to **1.5K requests/second** at **80–100 ms** latency.
- Implemented multi-cluster failover with OpenTelemetry, Prometheus, Grafana, ELK, and ClickHouse observability for **99.95% uptime**.

MeetQ | Software Engineer

AI-powered online meeting platform

- Built a real-time meeting backend on LiveKit integrating OpenAI models for transcription, voice translation, and summaries.
- Designed the real-time translation pipeline and optimized its conversational context handling for accuracy and coherence.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. Information Technology – Knowledge Engineering

DevOps on AWS Specialization | Coursera